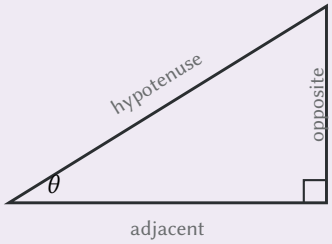
QR CODE **WATCH THE VIDEO BEFORE YOU SOLVE** YouTube • Lesson 22

**RULE · Right-triangle ratios (SOH-CAH-TOA)**

$$\sin\theta = \frac{\text{opp}}{\text{hyp}} \qquad \cos\theta = \frac{\text{adj}}{\text{hyp}} \qquad \tan\theta = \frac{\text{opp}}{\text{adj}}$$




**SOH-CAH-TOA:** Sin = Opp/Hyp, Cos = Adj/Hyp, Tan = Opp/Adj. Ratios are unitless.

**RULE · Opposite & adjacent flip with the angle**

The same leg is opposite for one acute angle and adjacent for the other – redefine them each time the angle changes.

3-4-5 triangle, legs $BC = 3$, $AC = 4$, hyp $AB = 5$:

| | From $A$ | From $B$ |
|---|---|---|
| opposite | 3 | 4 |
| adjacent | 4 | 3 |

RULES SHEET – 2/2

**RULE · Solving for a missing side**

| Known + unknown | Use |
|---|---|
| hypotenuse + opposite | sin |
| hypotenuse + adjacent | cos |
| adjacent + opposite | tan |

Hyp $= 12$, angle $= 25°$, find opposite: $\sin 25° = \dfrac{\text{opp}}{12} \Longrightarrow \text{opp} = 12 \sin 25° \approx 5.07$

*Keep the calculator in degree mode – radian mode gives a plausible-looking but wrong answer with no warning.*

**RULE · Missing angle & complementary identity**

$$\sin\theta = \cos(90° - \theta)$$

To find the angle itself, apply the inverse function to the ratio – the ratio is not the angle.

$$\sin\theta = \frac{7}{25} = 0.28 \Longrightarrow \theta = \arcsin(0.28) \approx 16.3°$$

$\sin\theta = 0.6$ *means* $\theta = \arcsin(0.6) \approx 36.9°$ – *not* $0.6°$.